Assembly overview

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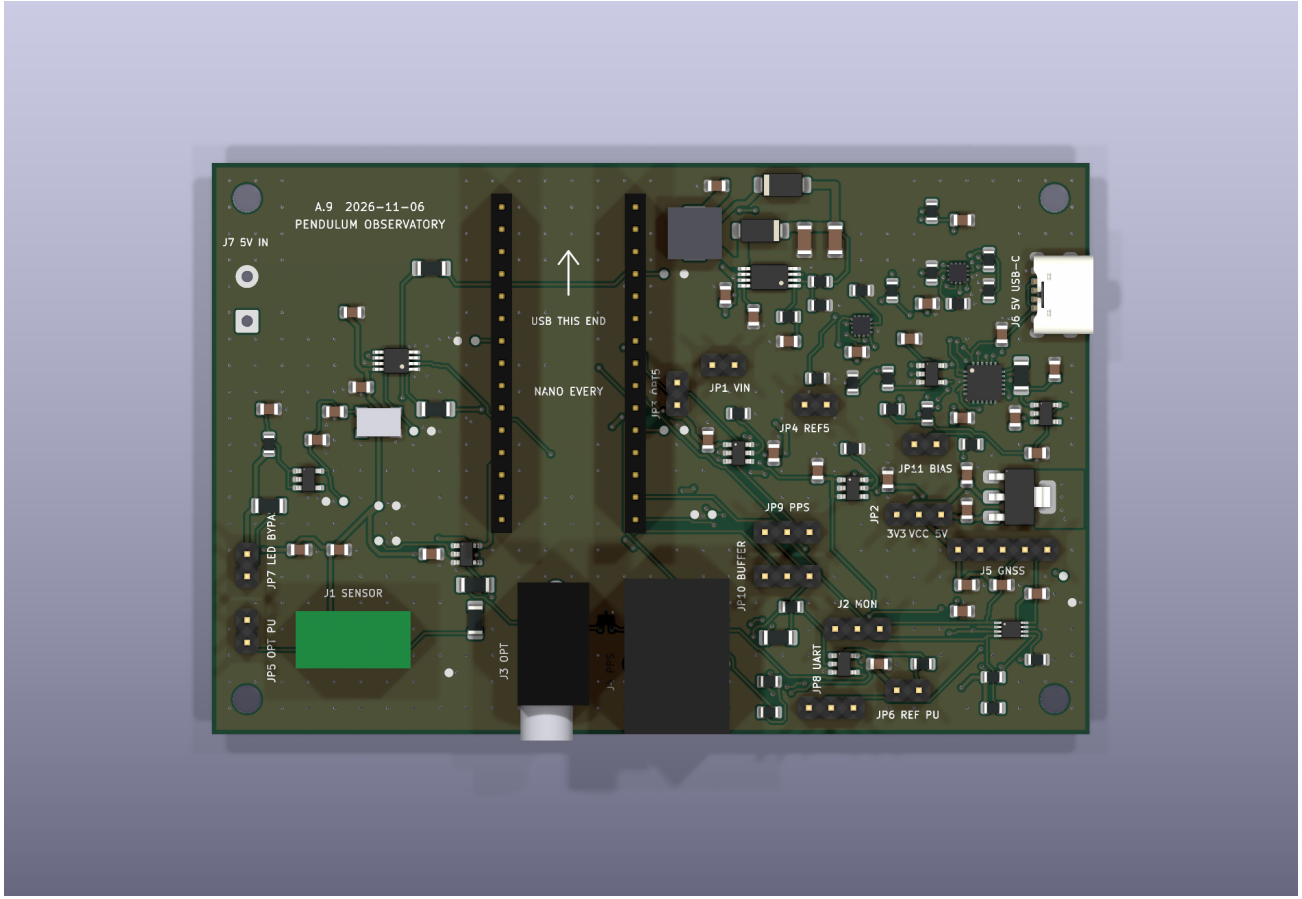
Stock Nano Every. Ordinary USB data cable. External power enters the Nano VIN header through the carrier boost stage. Do not modify the Nano or its Micro-USB cable.

105 fitted physical parts: 87 SMD/hybrid parts, including USB-C J6, and 18 through-hole pieces, including two A1 sockets. Optional J7 is one additional part. Test pads and mounting holes are not purchased parts.

Fit and inspect SMD parts first, then connectors/headers and two level socket strips. Keep the Nano and GNSS disconnected during first power checks. Read the exact BOM suffixes: the two TSSOP8 packages differ.

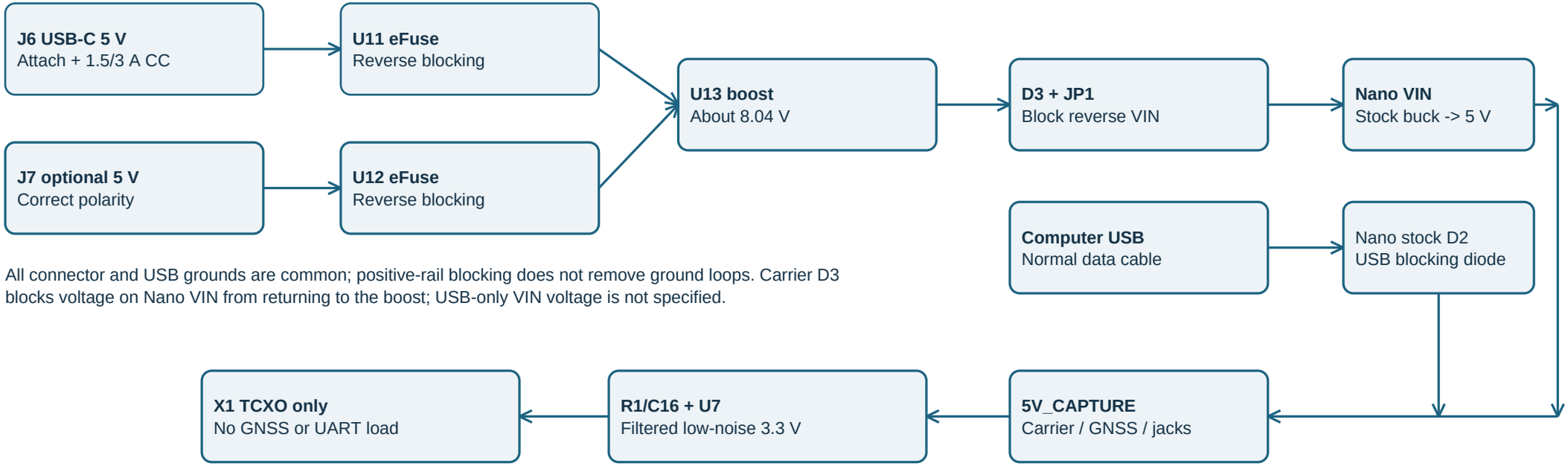
The arrow between the A1 rows points to the Nano USB end. Fit the Nano only after power checks. The inserted module covers the orientation label.

The small 3D envelopes for J1, L1 and the eFuses support placement review. Real manufacturer drawings and the delivered parts control fit. Optional J7 is hidden in the default render.



Power path and TCXO protection

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All connector and USB grounds are common; positive-rail blocking does not remove ground loops. Carrier D3 blocks voltage on Nano VIN from returning to the boost; USB-only VIN voltage is not specified.

Design quantity	Nominal result / acceptance boundary
External input	4.75-5.25 V; each eFuse about1 A. OVLO trip/reset about5.48/4.98 V: after a trip, disconnect, discharge and reconnect the source.
VIN conversion	8.04 V before D3; verify loaded Nano VIN remains within its recommended range.
USB fallback	USB feeds the 5 V rail through Nano D2, not VIN. Qualify host/cable/load: AHCT supplies >=4.5 V;4.75 V preferred.
TCXO filtering	22 ohm /10u input, dedicated TPS7A2033, local100n +4.7u output.
Load budget	Provisional 500 mA TOTAL Nano 5 V consumption under external power; conversion efficiency, startup and heat need bench checks.

Connectors and accepted inputs

Connector	Assignment / assembly requirement
A1	Two 1x15 female 2.54 mm sockets; stock Nano Every USB end points to arrow.
J1 screw terminal	1 LED+,2 GND,3 detector 5 V,4 detector signal,5 GND. Default Wurth five-way 2.54 mm.
J2 monitor	1 GND,2 raw GNSS PPS,3 receiver TX. Read-only; no supply or transmit connection.
J3 optical TRS	Tip optional 5 V OUT; ring signal IN; sleeve GND. Empty selects J1.
J4 external PPS TRS	Same tip/ring/sleeve. Insertion selects conditioned external PPS and drives Nano D3 LOW.
J5 GNSS	1 SELECTED VCC,2 GND,3 PPS,4 receiverTX,5 receiverRX. Common 2.54 mm header, not TRS. JP2 defaults to 3.3 V.
J6 / J7	J6 USB-C regulated 5 V input. J7 optional: pin 1 +5 V, pin 2 GND;5.08 mm, never9/12 V. No fixed source priority.

J3/J4 logic: nominal 3.3 V/5 V, working0-5.25 V; LOW <=0.4 V, HIGH >=2.2 V. Non-inverting AHCT Schmitt receivers. 100-ohm series input and ESD protection do not make these arbitrary-voltage inputs.

Pull-ups: selectable10k to 5 V. Use for a qualified open collector; source should sink 1 mA at the LOW limit. Leave open for 3.3 V push-pull or 3.3 V pulled-up outputs unless 5 V pull-up tolerance is established.

Power: jack tips are outputs. JP3/JP4 disconnect each tip. Provisional 75 mA combined tip load. Connect TRS leads with power off; an inserted dead source still overrides the original input.

PPS: one positive pulse per second, preferably100ms HIGH; rising edge is marker. Raw radio timecode and arbitrary polarities need separate qualification.

Jumper defaults

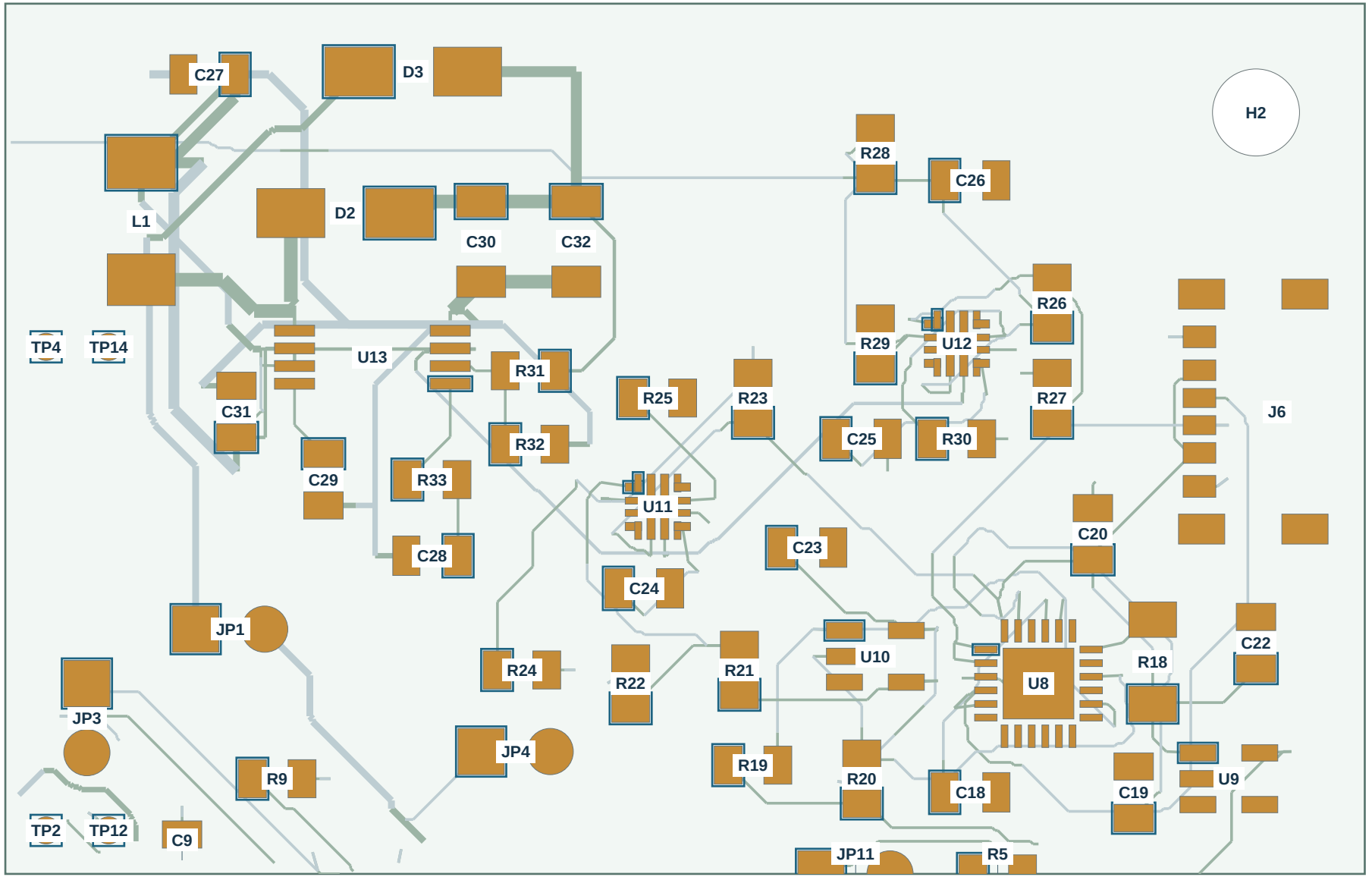
Header	Purpose	Assembly default / change
JP1	External VIN feed	FITTED. Open disconnects external feed; Nano USB can still power the board.
JP2	GNSS VCC selector	1-2=3.3 V DEFAULT; 2-3=5 V; removed disconnects J5 power. One shunt only.
JP3	J3 optical tip 5 V	FITTED. Open for an independently powered adapter.
JP4	J4 external tip 5 V	OPEN until the reference is qualified for 5 V power.
JP5	Optical10k pull-up	FITTED for installed OPB912. Open for an unsuitable push-pull/own-pull-up source.
JP6	External PPS10k pull-up	OPEN. Fit only for a suitable open-collector source.
JP7	LED resistor bypass	FITTED only with the existing145.5-ohm resistor-equipped lead. OPEN for a bare LED.
JP8	Nano UART	1-2 ENABLED; 2-3 disabled for receiver service.
JP9 + JP10	PPS comparison	BOTH 2-3 BUFFERED. Move BOTH to1-2 for direct comparison only.
JP11	GNSS PPS10k pull-down	OPEN. Qualify boot-shared receiver pin before fitting. Park spare on pin 2 only.

Eight bridging shunts: JP1/3/5/7, JP2 1-2 (3.3 V), JP8 1-2, JP9/10 2-3. Supply three spare shunts; one may park on a single JP11 pin. Never bridge all three pins or change settings under power.

Direct 3.3 V PPS needs independent Nano HIGH-threshold qualification. U1 is an ordinary AHCT buffer, not a Schmitt receiver. A.9 adds no hardware inversion and no new polarity firmware.

Power-section placement - component side

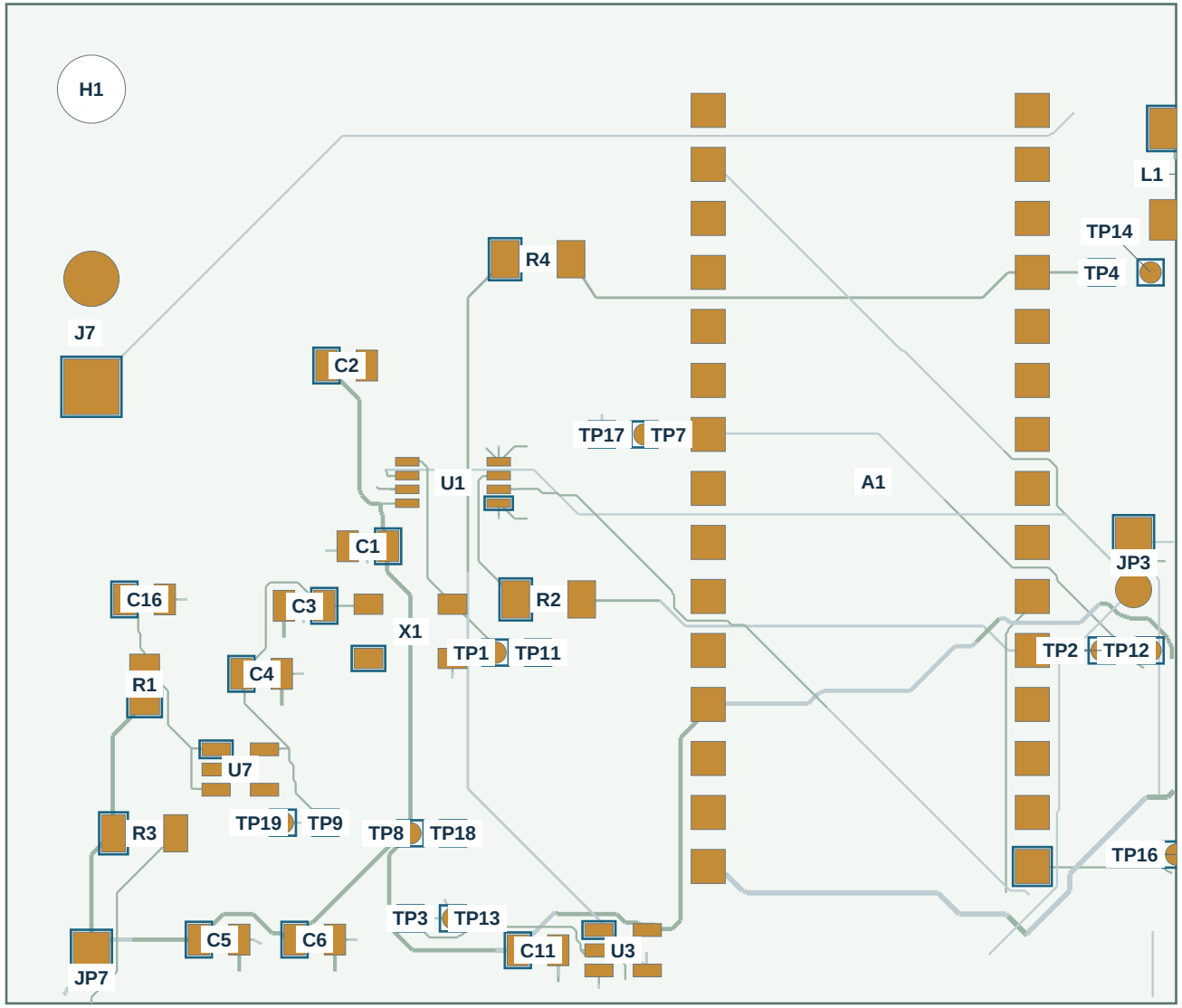
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Power off before assembly, TRS cable connection or shunt changes. Confirm delivered parts and pin 1.

Clock and quiet-supply placement

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Blue pad outline identifies pin 1. Pad numbers, package orientation and the native PCB control; leader labels identify components. No mapping assumes a plug rear view.

U7 supply: Nano 5 V ->R1 22 ohm ->C16/U7 ->3V3_TCXO. C3/C4 are local bypass. X1 supply has no accessory connection.

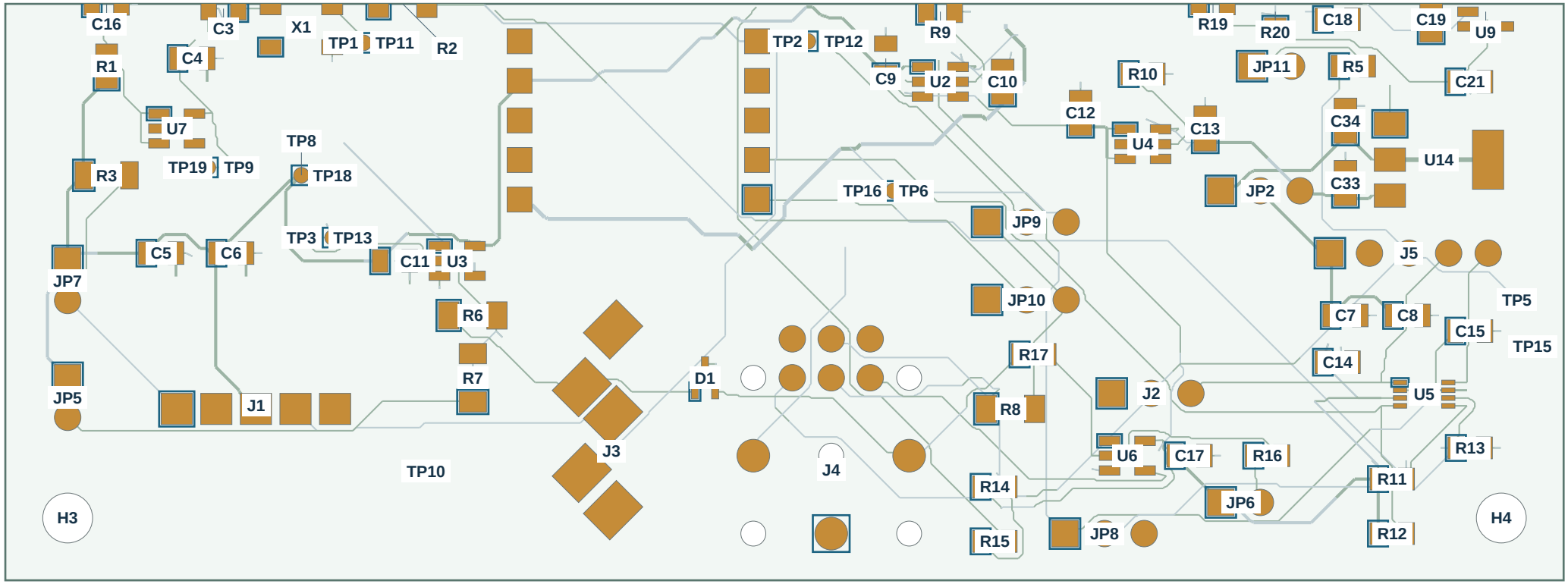
Clock path: X1 ->U1 clock gate ->R2 ->Nano D2. X1/U1/R2/Nano placement and post-R2 copper are preserved from A.8.

Measure at X1: loaded 3.3 V, ripple/spikes,16 MHz amplitude, duty cycle, jitter and startup. Use short probe grounds and state bandwidth.

TP1 raw clock;TP2 Nano clock;TP8 Nano 5 V;TP9 TCXO3V3;TP10 and board-onlyTP11-19 are ground. No purchased probe components.

Signal interfaces - component side

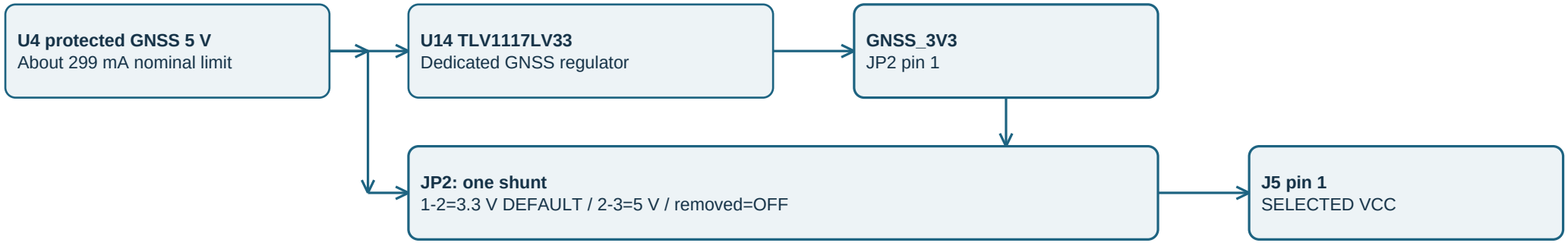
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Path	Physical and electrical check
Optical	J1 signal -> J3 normally closed ring ->R6/U3 ->R4 ->Nano D9. Plug ring replaces J1.
PPS	GNSS PPS -> J4 NC; external ring ->R17/U6 -> J4 NO. Selected common ->R8 ->JP9/10 ->NanoA3.
UART	J5/4 receiverTX -> level translator ->Nano D0/RX. Nano D1/TX -> translator ->J5/5 receiverRX. J2 is monitor only.

GNSS voltage selector and breadboard

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Header, not TRS: J5 uses five numbered 150-200 mm female-to-male 2.54 mm hobby leads to a 170-point breadboard. Verify the actual receiver pin map and voltage before connection; PPS/UART remain nominal 3.3 V.

JP2 pins: 1 GNSS_3V3; 2 GNSS_VCC to J5/1 and C7/C8; 3 GNSS_5V_LIMITED. Fit one two-pin shunt. A single proper shunt cannot connect both rails. All equipment, including receiver USB, must be off before changes.

Independent power: U14 has an OUT-to-IN diode and no reverse-current protection. Remove JP2 for an independently powered receiver. PPS/UART can still back-power an unpowered receiver; follow its service/isolation requirements.

U14 and capacitors: pin 1 GND, pin 2 AND tab OUT, pin 3 IN. Exact TLV1117LV33DCYR; C33 input 1 uF/16 V, C34 output 4.7 uF/16 V. Effective output capacitance must exceed 0.5 uF after derating. Keep receiver bypass at its power pins.

Load and heat: 200 mA provisional receiver-plus-active-antenna allowance. U14 loss is 0.34 W at 5 V or 0.39 W at 5.25 V. Verify enclosure temperature and loaded startup voltage at the receiver. Both choices follow U4; never power GNSS from TCXO or Nano3V3.

Mechanical support: adhesive-mount and insulate the breadboard; support the receiver and clamp its antenna cable separately. Record receiver model, selected voltage, pin map, antenna current and UART/PPS levels.

Bench acceptance - required before recording

Check	Evidence to record
Unpowered inspection	Exact MPN/package, diode stripe, ICpin1, terminal polarity, USB-C stake fit, socket orientation, shunts, shorts. Verify J3/J4 contacts with a real plug.
First external power	Nano/receiver removed. Current-limited source. Measure common 5 V input, boost output and JP1 VIN feed; then insert Nano and repeat at load.
Loaded rails and heat	Target5V_CAPTURE4.75-5.25 V, TCXO 3.3 V +/-2%. Check both JP2 voltages and removed setting, receiver startup, U14 heat at 200 mA, converter stability, current limiting and OVLO disconnect/discharge/reconnect recovery at 5.25 V.
TCXO noise	Proposed target <=1mV RMS 10 Hz-100 kHz, <=10mV peak-to-peak switching spikes at X1. State bandwidth/probe conditions; targets are not measured results. Check U7 shutdown reverse current against the 15 mA short-duration limit.
USB-C qualification	Both cable orientations;1.5A/3A enable, default-current/unattached reject, slow rise. Check no unintended turn-on during logic startup.
Backfeed / power changes	External only, USB only, both inputs and every removal sequence. Measure disconnected inputs, host VBUS leakage and USB-only Nano VIN. Verify D3 reverse blocking. Scope rails and clock with intended common-ground connections.
Signals and timing	Measure HIGH/LOW, slew, pulse widths, source selection, D3 insertion detection, direct/buffered delay and jitter; test missing/double edges and reference loss.
Full capture run	Optical + GNSS/PPS + USB traffic together, antenna load included. Check continuity and frequency under supply/load changes; keep a signed measurement record.

USB fallback is conditional. The 500 mA budget applies to external power. USB-only rail voltage is VBUS minus the Nano diode and wiring drops: verify >=4.5 V at U1/U3/U6 under full load;4.75 V is preferred. Check host current allocation. Power the full instrument externally before connecting the computer for normal use.

Power transfer may cause a reset or timing discontinuity. A backed-up external supply and backed-up logger are the intended route to continuity; this board is not a battery UPS. Do not claim seamless precision recording until measured.

Validation and revision boundaries

Carrier CAD: native ERC0; PCB DRC0 violations and0 unconnected; schematic parity0. Independent checks cover every active/interface terminal, all passive physical pad sets, actual schematic packages, supply separation and the clock geometry. See validation reports for exact assertions (1854 at document build).

Integrated GNSS supply: U14 pins and output tab, C33/C34, JP2 and J5 power mapping are checked as carrier circuitry. Both voltage choices follow U4; the TCXO supply is separate. No second PCB or separate regulator archive is needed.

A.8 stays closed: 285 design/release files remain byte-identical to the recorded working-tree baseline. Finder metadata, KiCad local preferences and Python bytecode are excluded from design checks. A.9 source and output live in a sibling directory. Do not mix A.8 and A.9 manufacturing bundles.

No firmware or Pi application deployment: power conversion, USB-C qualification and reverse blocking are hardware functions. Capture pins remain D2 clock,D9 optical,A3 PPS. D3 indicates the external jack. GNSS configuration, reference qualification/source reporting and optical polarity features remain separate firmware work.

Design is unbuilt: no thermal, analogue, jitter, power-transfer or mechanical acceptance result is claimed. The TI boost compensation network is a starting design needing bench stability checks. Third-party receiver/antenna compatibility is not inferred from its chipset name.

Source documents: README,Power_Design_A9.md,Interfaces_A9.md,Pinout_Audit_A9.md,Quoting_A9.md and Rebuild_A9.md. The separate wired schematic has exactly two pages: SIGNAL and POWER, with identical component-pin connectivity. Exact BOM, positions, accessory list and validation are in the source bundle. The quotation ZIP contains only the four supplier inputs; no order has been placed.

Manufacturer references: Arduino Nano Every schematic/full pinout; TI TPS25947,TPS61085,TPS7A20,TPS709,SN74LVC1G02,TXU0202,TPS2553,TPD2E2U06,TLV1117LV; STUSB4500L; Nexperia AHCT buffers; GCT,Kycon,Same Sky,Wurth,Phoenix and ECS drawings. See the pinout audit for URLs and local-reference revision provenance.